$$g_t = \nabla f(y_{t-1})$$

$$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$$

$$z \leftarrow z(1 - \eta\lambda) - \eta \cdot g_t / (\sqrt{v} + \varepsilon)$$

$$c_t = 1/(t+1)$$

$$x \leftarrow (1-c_t)x + c_t z$$

$$y_t = (1-\beta_1)z + \beta_1 x$$

carry y

y_t becomes y_{t-1}
(next step)